The Rare-Earth Vultures Harvesting Metals From America's Old Wind Turbines; Recycling ventures are part of a drive to loosen China's chokehold on the critical minerals

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FULL TEXT

To MidAmerican Energy, the wind turbines being taken down and dismantled at its Highland wind farm in O'Brien County, Iowa, are junk. To Brian Diesselhorst, they are the mother lode.

Diesselhorst is chief executive of Paladin EnviroTech, a recycling company eyeing a new role as a supplier of rare-earth metals. The U.S. is trying to reduce its reliance on China for rare earths, which go into everything from drones to MRI scanners. Paladin is among a few ventures betting that the quickest route to new supply goes through the junkyard.

Rare-earth metals with names like dysprosium and neodymium are used in magnets that go into electric motors. Diesselhorst is in the market for various expired gadgets. But retired wind turbines, whose room-size generators in some cases house huge rare-earth magnets, are his No. 1 target. Some more recent models can contain nearly a ton of rare earths, worth close to $100,000, Paladin estimates.

There are thousands of retirement-age turbines waiting to be harvested as America's wind industry ages. Only a fraction of the earlier models contain rare earths—20% of the turbines being replaced at the Highland facility have them—but they became commonplace in more-powerful turbines. Those are now starting to be replaced, so this source of waste is gradually becoming more valuable.

"That's why it was critical for us to get into the game now," said Diesselhorst, who says he believes Paladin is the first company to start harvesting this source of rare earths.

China established its rare-earth dominance over decades by providing financial support for domestic companies and flooding the market to squash nascent competitors. Last year, with the trade war with the U.S. heating up, Beijing flexed its muscles by imposing export restrictions that laid bare Western manufacturers' dependency on China.

Recycling ventures are one part of a drive to loosen that chokehold. The Trump administration is providing billions of dollars for mineral ventures, but the investments will take years to pay off.

For the most part, the old turbines being taken out of service are replaced with better ones. Those at the Highland wind farm have been spinning for a decade and could serve another decade, according to Adam Jablonski, vice president of project development at MidAmerican Energy, the Berkshire Hathaway-owned utility that owns the site.

But replacing them with better models, with blades 20% longer, will let each supply about 15% more power, Jablonski said—a boon when electricity demand is rising and developing new energy projects is slow and expensive.

MidAmerican has turned two things the Midwest has in abundance—wind and space—into the largest wind-power business of any U.S. utility. It has already replaced nearly 1,500 of its 3,700 turbines. Metal recyclers harvested what they could, but until Paladin came along the rare-earth metals that some of them contained were going to waste.

A barrier to entry is the cumbersome task of dismantling turbines. Cranes have to remove the blades and the nacelles—the room-size housings that contain the turbine generators. Then the components are sliced up with hydraulic shears and other equipment and loaded onto heavy-duty trucks.

"You're dealing with 80-to-120,000-pound pieces that you've got to load on flatbeds," said Luke Wray, the Paladin executive charged with overseeing the company's supplies of rare earths and other critical minerals. "You're on a field in the middle of

nowhere. If you've got mud, or you're on clay, and the groundsoil isn't sturdy, you could have a crane tip over."

On one job, when contractors were using explosives to fell turbines, hubs weighing more than 30 tons ended up buried deep in clay and had to be arduously excavated. Another risk comes from the wind itself—strong gusts can play havoc with blades being moved by cranes.

The other deterrent is the challenge of separating materials cost-effectively so the sought-after rare earths can be isolated, ground up and turned into metal for new magnets. Paladin works with Critical Materials Recycling, a Boone, Iowa company that is commercializing a leaching process developed at Ames National Laboratory to extract nearly all the rare earths from electronic waste.

Recyclers primarily targeted copper. Turbines contain several tons of copper in their wiring and other components, with a ton worth roughly $14,500 at today's prices.

Rare earths, harder to separate and less valuable, are generally recycled alongside steel and end up irretrievably lost in a byproduct called slag.

Several U.S. battery-recycling ventures ran into trouble in recent years in part because they couldn't source enough feedstock and bring costs down.

Diesselhorst is counting on securing enough turbines to run the recycling operation efficiently. Next year, Paladin forecasts, wind-turbine magnets will account for roughly 40% of its tonnage, with MRI machines and EV motors contributing about 25% each and hard drives accounting for the last 10%.

Wray has lined up a pipeline of about 500 turbines across the U.S. In the company's Tampa, Fla., offices, a poster of Don Quixote, the fictional knight who tilted at windmills, hangs on the wall, with a smaller portrait of Wray in the corner.

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DETAILS

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